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Paper Code : PEC-IT601D Image Processing

UPID : 006591

Time Allotted : 3 Hours *Full Marks :70*

The Figures in the margin indicate full marks.
Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following : [1 x 10 = 10]

(I) In image processing, _____ is the process of shifting an image from one location to another without altering its shape or size.

(II) Name the three types of distance measures used in image processing.

(III) The spatial domain method operates on the _____ values of an image.

(IV) _____ transformation is used to Rotate, scale, and warp an image.

(V) Which techniques is used to detect sharp changes in intensity in an image?

(VI) Image acquisition involves capturing an image using a _____.

(VII) The transformation that changes the size of an object while maintaining its shape is called _____.

(VIII) The ____ transform is mainly used in JPEG image compression.

(IX) _____ filtering is used to enhance edges and high-frequency details in an image.

(X) What does geometric transformation in image processing involve?

(XI) Isolated point can be best detected using which derivative?

(XII) The communication of images involves _____ from one location to another.

Group-B (Short Answer Type Question)

Answer *any three* of the following : [5 x 3 = 15]

2. Define Image Segmentation .What is the usefulness of segmentation? [5]

3. Describe the fundamental Steps in image processing. [5]

4. What is quantization and how does it affect the quality of a digital image? [5]

5. Derive the Euclidean, City-block and Chessboard distance formulas and compare their applications in image analysis. [5]

6. Perform histogram Stretching for the image distributions shown in the following table so that new image has dynamic range of [0,7]. [5]

Gray Level(r)	0	1	2	3	4	5	6	7
No. of Pixels(p)	0	0	50	60	50	20	10	0

Group-C (Long Answer Type Question)

Answer *any three* of the following : [15 x 3 = 45]

7. (a) What Do you mean by Image Negative? Explain. [5]

(b) If all the pixels of an image are shuffled, will there be any change in the histogram of that image? Justify your answer. [5]

(c) Explain Intensity Slicing with Example. [5]

8. (a) Discuss the importance of illumination-reflectance separation in homomorphic filtering. [5]

(b) Discuss the role of Point Spread Function (PSF) in image degradation and restoration. Provide practical examples. [10]

9. (a) Explain Edge detection using gradient operator. [5]

(b) Explain Edge linking using hough transform. [5]

(c) Explain the process of region extraction based on segmentation. [5]

10. How does digital image processing contribute to fields like medical imaging, remote sensing, and machine vision? Provide examples. [15]

11. (a) Compare unconstrained and constrained restoration methods. When is constrained restoration preferred? [5]

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(b) What is homomorphic filtering, and how does it help in image restoration? Explain with an example. [10]

*** END OF PAPER ***

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